

MICROSOFT INTERVIEW EXPERIENCE

Microsoft Corporation is an American multinational corporation and technology company headquartered in Redmond, Washington. Its best-known software products are the Windows line of operating systems, the Microsoft 365 suite of productivity applications, the Azure cloud computing platform and the Edge web browser. Its flagship hardware products are the Xbox video game consoles and the Microsoft Surface lineup of touchscreen personal computers.

Role Offered: Software Engineer

Stipend: 1,25,000 INR per Month

Process: Online Test, Technical Interview (2 Rounds)

Online Test:

It consists of two problems.

First Problem:

Given two numbers in the format of two strings. You can swap the corresponding digits of each number. The two numbers are of same length. You have to find the minimum swaps to make the absolute difference between them smaller.

Example: num1 = "2999" and num2 = "1000".

Explanation: The answer for this question is 1. Because if we swap the first digit of each number the numbers become 1999 and 2000. Their absolute difference between them becomes small. And also keep in mind that the swaps should be small. This question can be solved using the greedy approach.

Second Problem:

The second question is something special. This question can be solved in many ways. You are given a string and you are at infinite 2d space at a particular point and the string given says the direction in which you move in that space and you are allowed to move in all direction. But the given direction would contain all the direction except south. You have to move back to the start point with the minimum distance. And each direction given can be considered as a distance. The answer is the minimum path to return to the same point. (We should not touch the same path).

Example: S = "NEEEN", Answer = "ESSWWWW"

Explanation: The answer can be solved by the considering all the path as a grid and considering the path in which we came as a obstacle and the start point as the same point which we started and the endpoint we can compute.

ONLINE TEST EXPERIENCE:

Getting Selected for the next round is unbelievable and I thought it was a fake thing. I did my first round not that much. We a group of 5 members are selected for second round. I had a friend who got also selected from my batch we code together always so I was happy in the next process. I

started preparation 1 day night before interview. Actually, I can't believe that I was selected it was like dream. (I pinched me a lot :))

Before day preparation

The thing which got me selected. I saw about the technical interview conducted in various Companies especially everything which I want to know before the interview was given in this video I suggest everyone to watch the video if you are giving the technical interview in online(<https://youtu.be/MFkqnGiCCiU?si=6ZbsMmAzkwiQwfXc>)

Also referred the striver's standout video:
https://youtu.be/PG4t_6qjxz0?si=oRSF4ZqF5u7721g2

I was having some doubts about the OOPS Concept in c++ so referred this video by Shraddha Didi which I would recommend for cpp coders(<https://youtu.be/mlIUKyZIUUU?si=O8I2ocJs5YxAH-A4>)

The important and the most needed part is the resume questions preparation I was at dream I didn't believe I got selected in Microsoft so I didn't prepare well so I thought of an easy way for preparing the resume questions so there was a web developer friend from my batch who have prior experience for this type of companies so I asked him to ask me questions. he asked me the questions which came to my interview this is a reason I got selected in my interview the same questions can you believe it even I can't.

First Round:

The next Day morning I was in the state how I will get Microsoft and we are waiting for the link and we thought it won't come but at a sudden it came at 9.30 and we completed interview at the same time. They asked me two questions

- 1) No of subarrays with a given sum
- 2) Insert an element into binary search tree.

The questions were easy and I was talking continuously to the interviewer so that he didn't get bored and he spoke well to me. As the questions were easy, I was having extra time for about 15mins left in 45mins so he asked me about my projects and the questions which I discussed with my friends last night.

They didn't declare the result for the round 1 but they shortlisted for second round we a group of three gave the second round it was a round with a big question (Medium Level).

Second Round:

My question was given a vector of strings and a string prefix and I have to return all the strings which has the given prefix

Example: v = {"apple", "applet", "aptitude", "banana"} prefix = "app" result = {"apple", "applet"}

The question seems to be simple. You can solve it in $O(n)$ is the first thought for everyone but the interviewer asked me to sort the strings as consider it as a google search so sort it he said so we had an argument how $n \log n$ is faster than n for about 15 mins in which I had only 45mins for my solving part, then I wrote the solution $n \log n$ by sorting he was not satisfied he asked me to optimize the solution so I optimized it by finding the starting point in which I want to start search by using binary search in $\log n$. Now my solution is $n \log n + \log n$ then the interviewer asked me to bucketize the words so the I understood he was trying to make me code the bucket sort but I don't know bucket sort so I said that I know bucket sort exists but I don't know how it is implemented we didn't have time to code it as well so the meeting ended soon, I didn't code bucket sort but after that

I was waiting for the round 3 interview but the results were declared I cant believe that my name came still it is like in dream.

Points to be Remembered:

Luck plays a lot of role in internships, If you gets selected for the first round interview after online die hard to get selected as many people will be there for the competition if you go back.

Life takes you to the place which is for you prepare well and contact me anytime.

Thank You

Deepan B,

CEG CSE'26

Email: bkavin04@gmail.com

Mobile : +91 8072790498